

ISLAND EVOLUTION

Many rare and unusual birds live on islands, such as the Hawaiian Islands (shown below), the Galápagos Islands, Madagascar, and New Zealand. These species have developed in unique ways because they have been cut off from their relatives on the mainland for a long time. For example, on the Hawaiian Islands, a finch-like bird arrived some 5–6 million years ago. Since there were few other birds to compete with it, this bird evolved into around 40 different species called honeycreepers.



Mountain habitats

Mountains such as the Himalayas in Asia (right), the Andes in South America, the Rocky Mountains in North America, and the Alps in Europe provide a wide range of bird habitats. The lower slopes have warm forests, but these merge into grasslands and tundra higher up. Above a certain height—called the tree line—there are no trees because it is too cold for them to grow. Near the very top, the ground is covered with snow and ice, and no birds can live there. Mountain birds have to cope with freezing temperatures, fierce winds, and thin air. Some birds move up and down the mountains with the seasons.



Scrublands

This warm, dry, dusty habitat with tough shrubs and small trees is found mainly around the shores of the Mediterranean Sea, California in the US, and parts of Australia, where it is called the outback or the bush. Many birds move, or migrate, to these scrublands during the long, hot summers, when there are plenty of insects and seeds to feed on. Some fly away for the cooler, wetter winter months.

Find out more: pages 30–31, 52–53



Deserts

Deserts cover about one-fifth of the Earth's land surface. They are a difficult habitat for birds because there is little rainfall and daytime temperatures are very high. Birds have to rest in the shade during the hottest times of the day and either get water from their food or fly long distances to find it. Important deserts of the world include the western deserts of North America; the Sahara, Kalahari, and Namib deserts in Africa; and the Australian deserts.

Find out more: pages 15, 52–53



Rainforests

Rainforests grow near the equator where it is hot and wet all year. They cover less than 10 percent of the Earth's surface but are home to more than half of all the different species of wildlife living on Earth. Many birds live high up in the treetops, where there is more sunlight, warmth, and food. But birds live at all levels in the forest, sharing the food and space of this rich habitat. The largest birds live on the forest floor.

Find out more: pages 20–21, 36–37, 54–55



Wetlands

Watery environments are a favorite habitat for birds. Herons, ducks, geese, and swans flock to wetlands because there are plenty of fish, insects, and water plants for them to eat, as well as reeds and riverbanks for them to nest in. Trees grow in swampy areas, but there are hardly any trees on marshland. Drainage and pollution of their wetland habitats by people is a major threat to many of these birds.

Find out more: pages 16–17, 40–41

Living together

In any one habitat, different species of bird live side by side. In a rich habitat, such as a deciduous woodland, various kinds of birds can live on a single tree by feeding and nesting at different levels and eating different types of food. Some eat insects, while others prefer seeds. In this way, the birds share the resources of the habitat and are more likely to survive than if they compete with each other for the same things.

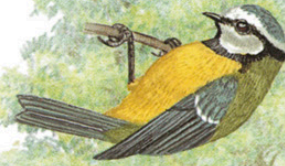
Near the top of the tree, tiny birds such as blue tits and wood warblers hang from the smaller twigs and pick insects off the leaves and bark.

In the middle of the tree, birds such as spotted flycatchers dart out from a perch to catch flying insects. Woodpeckers chisel into trunks and branches to find insects. They also dig out nesting holes in the trunk.

On the woodland floor, bigger birds such as the woodcock feed and nest. These birds are usually well camouflaged among the dead leaves. Wrens and other small insect-eaters hop nimbly through the dense thickets of leaves and twigs, where they are hidden from predators.



Wood warbler
(*Phylloscopus sibilatrix*)



Eurasian blue tit
(*Cyanistes caeruleus*)



Great spotted woodpecker
(*Dendrocopus major*)



Spotted flycatcher
(*Muscicapa striata*)

Eurasian woodcock
(*Scolopax rusticola*)



Northern wren
(*Troglodytes troglodytes*)

